

CODE

You just need the latest version of neuron, python and numpy. You also need to install neurotools which can easily be done with pip install NeuroTools. But you must replace stgen.py with the stgen.py file provided inside the “For_Werner” folder because a lot of the methods they used in the script are deprecated. It’s also assumed that you have the modules “time” and “datetime” with the latest version of python.

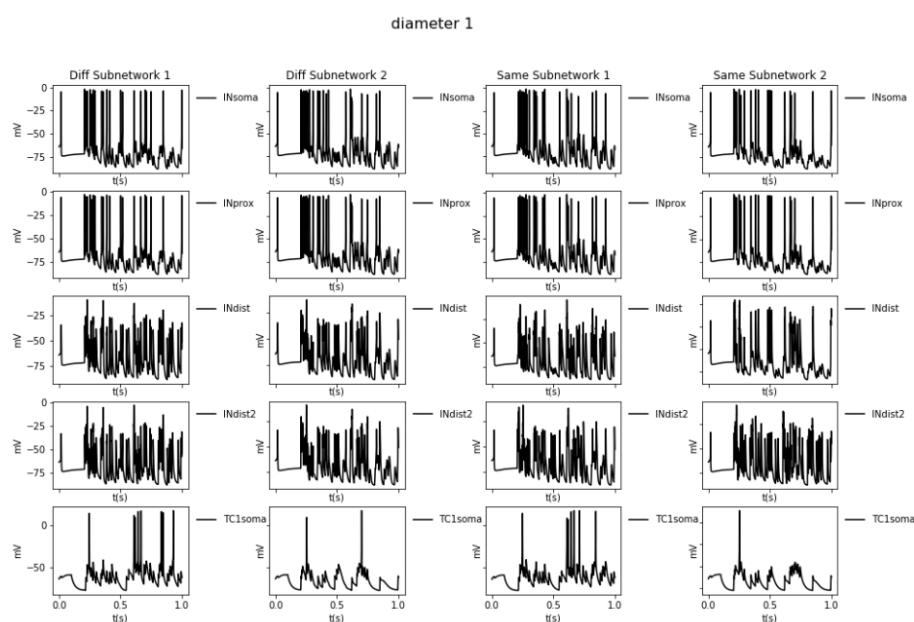
All the code is provided as .py files in the memodels/mechanisms folder:

cell_parallel.py - class objects of individual cells are created
subnetwork_parallel.py - class objects of subnetworks are created
poisson_parallel.py - creates poisson input spikes and saves them
diffrun_parallel.py - subnetworks are interconnected as diffIN arrangement and simulation is run
samerun_parallel.py - like diffrun, but subnetworks are interconnected in sameIN arrangement

here is the same code without parallel context that works just fine in serial:

cell.py
subnetwork.py
poisson.py
diffun.py
samerun.py

To run the serial/parallel versions, just compile the mod files located inside the memodels/mechanisms folder and run “python diffun.py” and then run “python samerun.py” (for parallel: “mpiexec -n 2 python diffun_parallel.py” and then “mpiexec -n 2 python samerun_parallel.py”). Each should take less than 5 minutes. (there are already some input spike files ready, or you can just rerun poisson.py and it will give you new randomly generated poisson spikes for subnetwork1 and subnetwork2, then run diffun.py and samerun.py). There is a python notebook in the spike_output folder called Analysis.ipynb where you can plot the spikes (you just need to change the name of the .txt or .npy file that’s saved after the simulation runs) and it should look exactly like this unless you made new random poisson spike inputs:

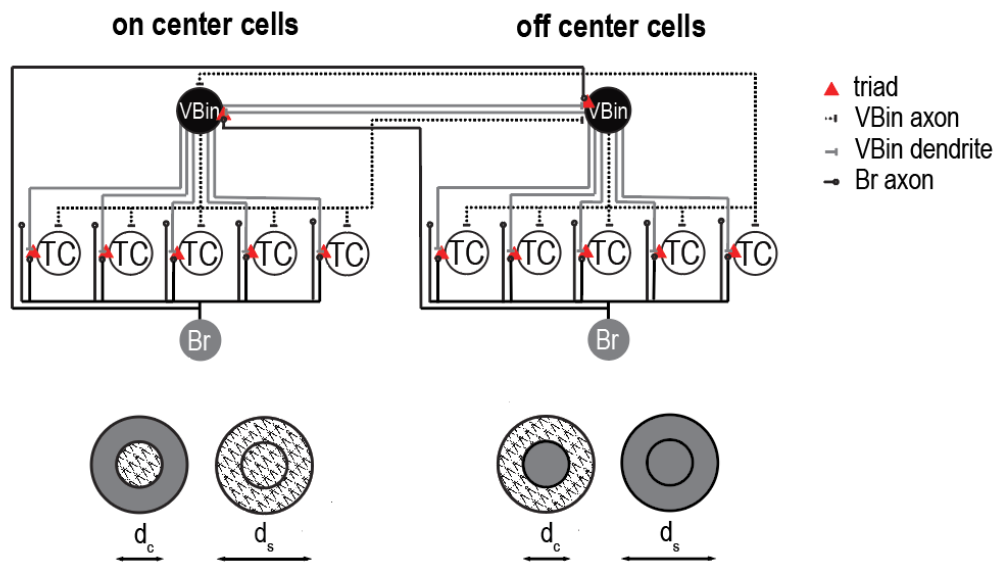


The rest of this document explains the network and spike inputs, it's not necessary to read these unless you're really getting into the details of the code

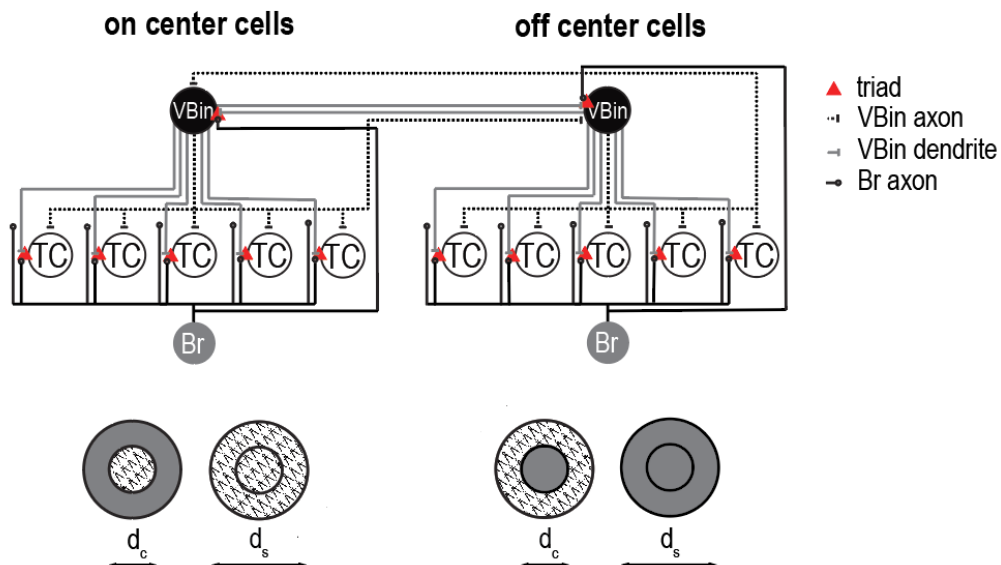
GOAL AND EXPLANATION OF NETWORK

My goal is to see how the response of the neurons change by switching the subnetwork connectivity from: Br (Brainstem) to different sign VB (Ventrobasal) Interneuron, to, Br (Brainstem) to same sign VB (Ventrobasal) interneuron. ON center and OFF center just represent the sign of the receptive field properties of the neurons in each subnetwork. This just means the neurons of each subnetwork respond to opposite stimuli (signs). My research question is what is the point of VBin to VBin triadic disinhibition. It's a very weird type of synapse. My hypothesis is that the disinhibitory triad serves as a **gating mechanism**. There needs to be two coincident inputs from the brainstem (same sign) in order for the signal to pass. So the "sameIN" arrangement will allow for signals from the Br to pass to the interneuron of the same sign. Br and TC (thalamocortical relay) are excitatory, VBin (a.k.a IN) is inhibitory.

Br to different VBin (I call this the diffIN connectivity arrangement)

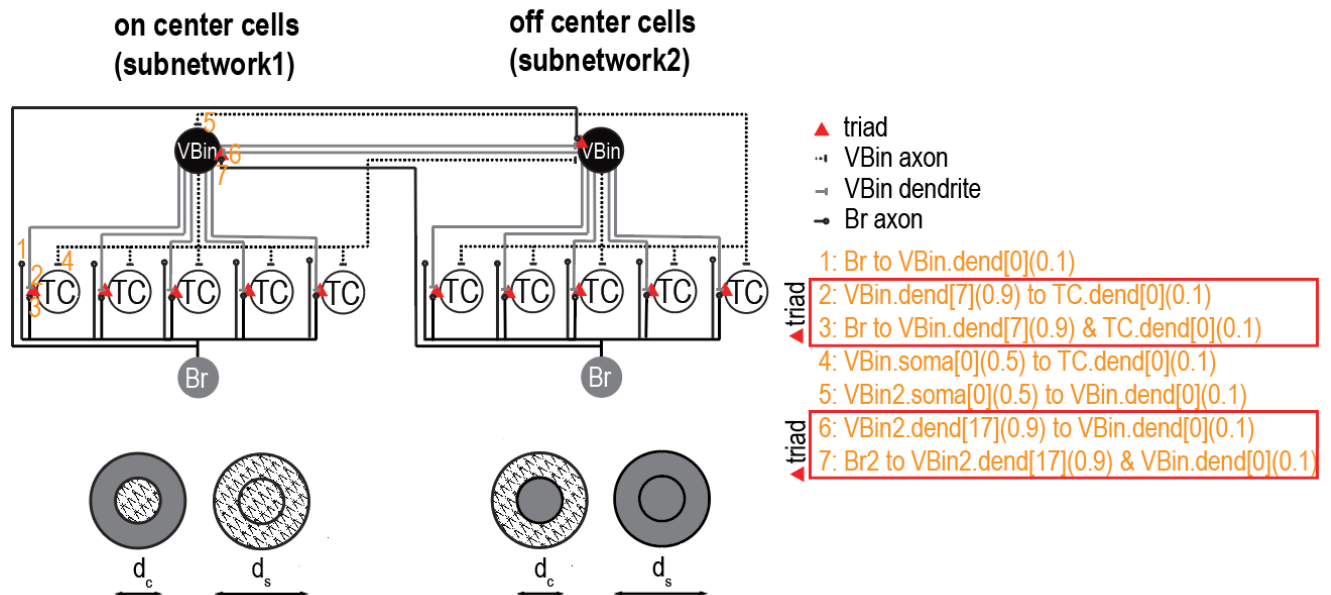


Br to same VBin (I call this the sameIN connectivity arrangement)

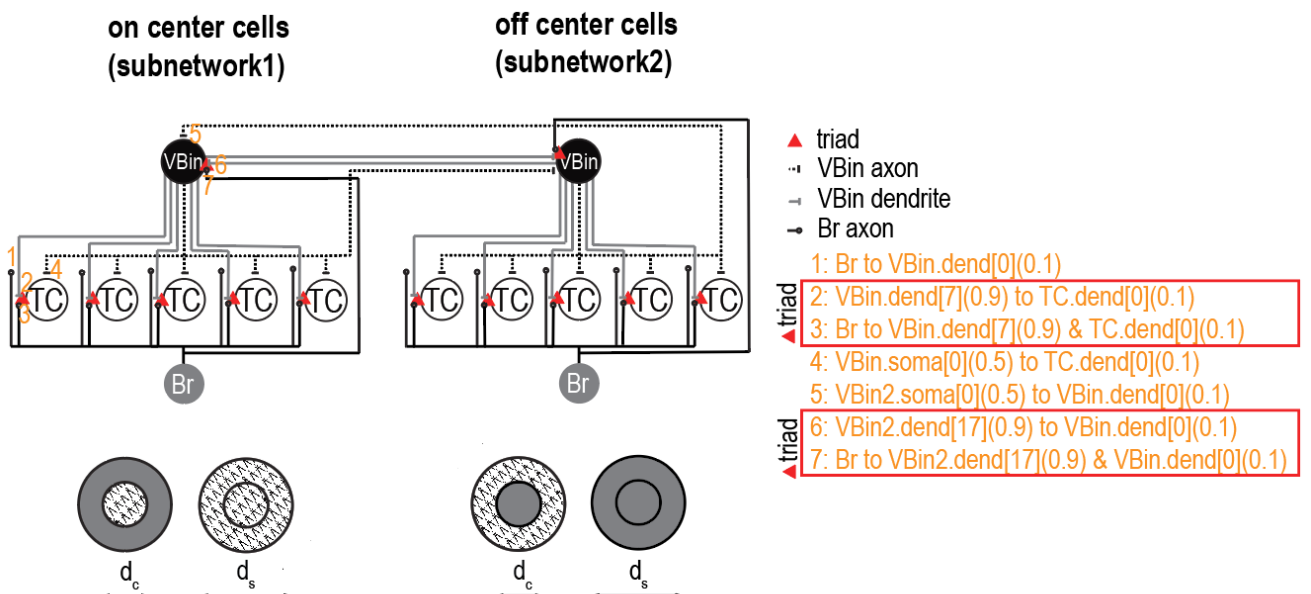


Here are the same subnetworks with all the NetCons shown.

diffIN arrangement



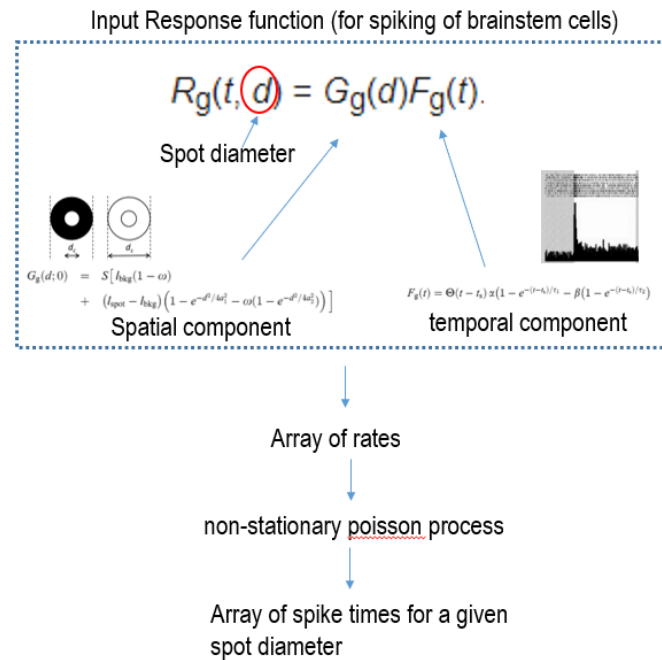
sameIN arrangement



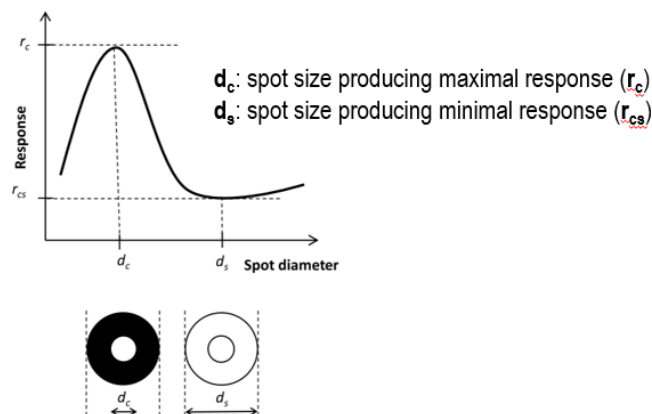
The interesting thing about this network is that it has dendro-dendritic synapses as well as triadic synapses. NOTE: here I'm showing 5 TC neurons per subnetwork, but for the moment in my test model, I'm only using 2 TC neurons since they are identical cells for the moment anyway.

INPUT and OUTPUT

The input (spiking of the Br cells) is defined here. Basically, it is a response function $R(t,d)$ where t is time and d is spot size diameter. Spot size diameter is just a stimulus spot given to an animal. $R(t,d)$ is defined by $G(d)$ the spatial component, and $F(t)$, the temporal component. The $G(d)$ depends on spot diameter such that there is an optimal spot size that corresponds to the receptive field of the cell. The temporal component $F(t)$ is added to create the “overshoot” in response. The input for subnetwork1 will be a 200ms delay + 400ms of full field (surround) stimulation (max d at ~ 10 degrees, this is basically “noise”) + 400ms of a different diameter ranging from 0-10 degrees. The input for subnetwork2 will be 200ms delay + 800ms of full field stimulation (“noise”).



After the simulation runs, I will gather the spiking data to see firing rate as a function of spot diameter and you can look at some interesting aspects about these values etc etc.



From these quantities, look at:

- d_c^{TC}/d_c^{Br} (change in RF size)
- $\alpha = (r_c - r_{cs})/r_c * 100\%$ (center-surround antagonism)

More info here: Biophysical Network Modelling of the dLGN Circuit: Different Effects of Triadic and Axonal Inhibition on Visual Responses of Relay Cells. Heiberg et al., 2016